

INF 1004: Mathematics 2 — Solutions

Tutorial : Matrix Algebra & Linear Systems

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Singapore Institute of Technology

Solution 1: Matrix Arithmetic

Given $A = \begin{pmatrix} 4 & 1 \\ 2 & 6 \end{pmatrix}$, $B = \begin{pmatrix} -3 & 5 \\ 0 & 9 \end{pmatrix}$, $C = \begin{pmatrix} 7 & 0 & 4 \\ 1 & 0 & -6 \end{pmatrix}$

- **(a) $A + B$ and $B + A$:**

$$\begin{pmatrix} 4 - 3 & 1 + 5 \\ 2 + 0 & 6 + 9 \end{pmatrix} = \begin{pmatrix} 1 & 6 \\ 2 & 15 \end{pmatrix}$$

Note: $A + B = B + A$ (Commutative Property).

- **(b) $A + C$: Undefined.** A is 2×2 and C is 2×3 .
Dimensions must match.
- **(c) $(C^T)^T$:**

$$C^T = \begin{pmatrix} 7 & 1 \\ 0 & 0 \\ 4 & -6 \end{pmatrix} \implies (C^T)^T = \begin{pmatrix} 7 & 0 & 4 \\ 1 & 0 & -6 \end{pmatrix} = C$$

Solution 2: Parameter Identification

Equating elements from: $\begin{pmatrix} a + c & 4 + 2 & 6 + 1 \\ b + 2 & 1 + 2 & 3 + d \end{pmatrix} = \begin{pmatrix} 5 & 6 & 7 \\ 9 & 3 & 4 \end{pmatrix}$

- $a + c = 5$
- $b + 2 = 9 \implies \mathbf{b = 7}$
- $3 + d = 4 \implies \mathbf{d = 1}$
- a, c can be any values summing to 5.

Note: Usually, we need a second equation to fix a and c uniquely.

Solution 3: Matrix Multiplication

$$A = \begin{pmatrix} 3 & 2 \\ 0 & -1 \\ 5 & 2 \end{pmatrix} (3 \times 2), B = \begin{pmatrix} 1 & 4 & 2 \\ -2 & 3 & 0 \end{pmatrix} (2 \times 3)$$

- **AB** (3×3):

$$\begin{pmatrix} 3(1) + 2(-2) & 3(4) + 2(3) & 3(2) + 2(0) \\ 0(1) - 1(-2) & 0(4) - 1(3) & 0(2) - 1(0) \\ 5(1) + 2(-2) & 5(4) + 2(3) & 5(2) + 2(0) \end{pmatrix} = \begin{pmatrix} -1 & 18 & 6 \\ 2 & -3 & 0 \\ 1 & 26 & 10 \end{pmatrix}$$

- **BA** (2×2):

$$\begin{pmatrix} 1(3) + 4(0) + 2(5) & 1(2) + 4(-1) + 2(2) \\ -2(3) + 3(0) + 0(5) & -2(2) + 3(-1) + 0(2) \end{pmatrix} = \begin{pmatrix} 13 & 2 \\ -6 & -7 \end{pmatrix}$$

Solution 4: Proof of Inverse

Target: Show $(AB)(B^{-1}A^{-1}) = I$.

1. Start with: $(AB)(B^{-1}A^{-1})$
2. Group middle terms (Associative Property): $A(BB^{-1})A^{-1}$
3. Use inverse definition $BB^{-1} = I$: $A(I)A^{-1}$
4. Identity property: AA^{-1}
5. Use inverse definition $AA^{-1} = I$: **I**

Since the product results in I , $B^{-1}A^{-1}$ is by definition the inverse of AB .

Solution 5: Solving for Matrix A

$$\text{Given: } [3A^T + \begin{pmatrix} 4 & 0 \\ 1 & 2 \end{pmatrix}]^T = A + \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}^{-1}$$

$$1. \text{ Apply transpose to LHS: } 3A + \begin{pmatrix} 4 & 1 \\ 0 & 2 \end{pmatrix} = A + \frac{1}{6} \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$$

$$2. \text{ Isolate } A \text{ terms: } 2A = \begin{pmatrix} 1/2 & 1/6 \\ 0 & 1/3 \end{pmatrix} - \begin{pmatrix} 4 & 1 \\ 0 & 2 \end{pmatrix}$$

$$3. 2A = \begin{pmatrix} -7/2 & -5/6 \\ 0 & -5/3 \end{pmatrix}$$

$$4. \mathbf{A} = \begin{pmatrix} -7/4 & -5/12 \\ 0 & -5/6 \end{pmatrix}$$

Solution 6: Inverse of Matrix B

1. **Determinant ($|B|$):** Expanding along Row 1:

$$|B| = 1(3 - 0) - 0(6 - 0) + 2(2 - 0) = 3 - 0 + 4 = 7$$

2. **Cofactor Matrix (C):** Using $C_{ij} = (-1)^{i+j} M_{ij}$:

$$M = \begin{bmatrix} 3 & 6 & 2 \\ -2 & 3 & 1 \\ -2 & -4 & 1 \end{bmatrix} \xrightarrow{\text{Apply Signs}} C = \begin{bmatrix} 3 & -6 & 2 \\ 2 & 3 & -1 \\ -2 & 4 & 1 \end{bmatrix}$$

3. **Adjoint Matrix ($\text{adj}(B)$):** Transpose of C :

$$\text{adj}(B) = C^T = \begin{bmatrix} 3 & 2 & -2 \\ -6 & 3 & 4 \\ 2 & -1 & 1 \end{bmatrix}$$

4. **Inverse Matrix (B^{-1}):** Using $B^{-1} = \frac{1}{|B|} \text{adj}(B)$:

$$B^{-1} = \frac{1}{7} \begin{bmatrix} 3 & 2 & -2 \\ -6 & 3 & 4 \\ 2 & -1 & 1 \end{bmatrix}$$

Solution 7: Proof and Decomposition

1. Proof of Zero Diagonal:

- For a skew-symmetric matrix, $m_{ji} = -m_{ij}$ for all i, j .
- For diagonal elements ($i = j$): $m_{ii} = -m_{ii}$.
- $2m_{ii} = 0 \implies m_{ii} = 0$.

2. Decomposition of $A = \begin{pmatrix} 6 & 8 \\ 2 & 10 \end{pmatrix}$:

- $S = \frac{1}{2} \left[\begin{pmatrix} 6 & 8 \\ 2 & 10 \end{pmatrix} + \begin{pmatrix} 6 & 2 \\ 8 & 10 \end{pmatrix} \right] = \frac{1}{2} \begin{pmatrix} 12 & 10 \\ 10 & 20 \end{pmatrix} = \begin{pmatrix} 6 & 5 \\ 5 & 10 \end{pmatrix}$
- $V = \frac{1}{2} \left[\begin{pmatrix} 6 & 8 \\ 2 & 10 \end{pmatrix} - \begin{pmatrix} 6 & 2 \\ 8 & 10 \end{pmatrix} \right] = \frac{1}{2} \begin{pmatrix} 0 & 6 \\ -6 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}$

Observe that $S + V = \begin{pmatrix} 6+0 & 5+3 \\ 5-3 & 10+0 \end{pmatrix} = \begin{pmatrix} 6 & 8 \\ 2 & 10 \end{pmatrix} = A$.

Verification: $S^T = \begin{pmatrix} 6 & 5 \\ 5 & 10 \end{pmatrix} = S$, $V^T = \begin{pmatrix} 0 & -3 \\ 3 & 0 \end{pmatrix} = -\begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix} = -V$